TD7: American Options - Step-by-Step Solutions

ST7 Stochastic Finance and Risk Modelling

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1 Model Setup and Value Function Properties

Question 1: Recursive Formulation of the American Option Value

Problem Statement: Prove that $V_t = v(t, S_t)$ such that:

$$\begin{aligned} v(T, x) &= h(x) \\ v(t-1, x) &= \max\left\{h(x), \tilde{p}v(t, x(1+d)) + \tilde{q}v(t, x(1+u))\right\}, \quad t = 1, \ldots, T \end{aligned}$$

where $\tilde{p}$ and $\tilde{q}$ are positive constants to be determined.

Step-by-Step Solution:

1. **Terminal Condition:** At maturity $t = T$, the option expires. The value of the American option is simply its intrinsic payoff. Thus, $V_T = h(S_T)$, which implies $v(T, x) = h(x)$.

2. **Dynamic Programming Principle:** At any time $t - 1$, the holder of an American option must choose between two mutually exclusive actions:

   - *Exercise immediately:* The payoff received right now is $h(S_{t-1})$.
   - *Hold the option (Continuation):* The option is kept alive until at least time t. Its value is the discounted expected value of the option at time t under the risk-neutral measure Q.

3. **Calculating the Continuation Value:** Because the market is complete and arbitrage-free, the Fundamental Theorem of Asset Pricing states that the expected discounted value under Q is:

   $$\frac{1}{1+r}\mathbb{E}_Q[V_t \mid \mathcal{F}_{t-1}]$$

   We know that $V_t = v(t, S_t)$. Under the binomial model, the asset price S_t can transition from S_{t-1} to:

   - $S_{t-1}(1+u)$ with risk-neutral probability $q = \frac{r-d}{u-d}$
   - $S_{t-1}(1+d)$ with risk-neutral probability $1 - q$

   Substituting these into the expectation gives the continuation value:

   $$\frac{1}{1+r}\left((1-q)v(t, S_{t-1}(1+d)) + qv(t, S_{t-1}(1+u))\right)$$

4. **Identifying the Constants:** Matching this expectation with the given formula yields:

   $$\begin{aligned} \tilde{p} &= \frac{1-q}{1+r} \\ \tilde{q} &= \frac{q}{1+r} \end{aligned}$$

   Given the problem condition $-1 < d < r < u$, it follows that $0 < q < 1$ and $1 + r > 0$. Therefore, both $\tilde{p}$ and $\tilde{q}$ are strictly positive.

5. **Conclusion:** The rational option holder will choose the action that maximizes their payoff. Taking the maximum of the immediate exercise value and the continuation value yields the desired recursive relation:

   $$v(t-1, x) = \max\left\{h(x), \tilde{p}v(t, x(1+d)) + \tilde{q}v(t, x(1+u))\right\}$$

Question 2: Shape Preservation of the Value Function

Problem Statement: If h is non-decreasing (resp. non-increasing, convex, continuous), then $x \mapsto v(t,x)$ is non-decreasing (resp. non-increasing, convex, continuous) for all $t = 0, \ldots, T$.

Step-by-Step Solution: We prove this by backward induction on time t.

1. **Base Case ($t = T$):** We have $v(T,x) = h(x)$. Since the payoff function $h(x)$ inherently possesses the property in question, $v(T,x)$ trivially holds the same property.

2. **Inductive Hypothesis:** Assume that for some time $t \leq T$, the function $x \mapsto v(t,x)$ possesses the required property (e.g., convexity). We must show that $x \mapsto v(t-1,x)$ also possesses it.

3. **Analyzing the Continuation Function:** Let us define the continuation value function $C(x)$ at time $t-1$:
$$C(x) = \tilde{p}v(t, x(1+d)) + \tilde{q}v(t, x(1+u))$$
   - From our base assumptions, $d > -1$, meaning $(1+d) > 0$ and $(1+u) > 0$. Scaling the input x by a positive constant preserves monotonicity, convexity, and continuity. Thus, the terms $v(t, x(1+d))$ and $v(t, x(1+u))$ maintain the property.
   - Since $\tilde{p} > 0$ and $\tilde{q} > 0$, taking a linear combination with strictly positive weights also preserves these properties. Therefore, $C(x)$ inherits the exact same shape properties as $v(t, \cdot)$.

4. **Applying Properties to the Maximum Operator:** Recall the recursive relationship:
$$v(t-1,x) = \max\{h(x), C(x)\}$$
   A fundamental property of the max operator is that it preserves these specific functional shapes:
   - The maximum of two non-decreasing functions is non-decreasing.
   - The maximum of two non-increasing functions is non-increasing.
   - The maximum of two convex functions is convex.
   - The maximum of two continuous functions is continuous.

   Since both the immediate payoff $h(x)$ and the continuation value $C(x)$ possess the required property, their maximum $v(t-1,x)$ does as well. This completes the inductive step, proving the statement for all $t = 0, \ldots, T$.

2 Positivity Conditions for Put Options

We now assume that the payoff function is specifically for a put option, meaning $h(x) = (K-x)_+$ for a given strike price $K > 0$. The initial price of the underlying asset is denoted as $S_0 = s_0 > 0$.

Question 3: Positivity of the European Put

Problem Statement: Show that $V_0^e > 0$ if and only if $s_0 < \frac{K}{(1+d)^T}$.

Step-by-Step Solution:

1. **Risk-Neutral Valuation:** As established in Question 1, the market is complete and arbitrage-free. By the Fundamental Theorem of Asset Pricing (FTAP), the initial value of the European option is the discounted expected payoff under the risk-neutral measure Q:
$$V_0^e = \mathbb{E}_Q\left[\frac{(K - S_T)_+}{S_T^0}\right]$$
   where $S_T^0 = (1+r)^T$ is the deterministic value of the riskless asset at maturity.

2. **Expectation of a Non-Negative Random Variable:** The payoff $(K-S_T)_+$ is strictly non-negative. A fundamental property of probability states that the expectation of a non-negative random variable is zero if and only if the random variable is zero almost surely (a.s.). Therefore:

$$V_0^e = 0 \iff (K - S_T)_+ = 0 \text{ holds } Q\text{-a.s.}$$

3. **Evaluating the Almost Sure Condition:** The condition $(K - S_T)_+ = 0$ is equivalent to saying $S_T \geq K$. For this to be true almost surely under Q, it must hold for the absolute minimum possible value of S_T.

4. **Minimum Possible Price:** In the binomial model, the lowest possible price at maturity occurs if the asset price goes down at every single step $t = 1, \ldots, T$. This minimum price is $s_0(1+d)^T$.

5. **Conclusion:** For the option value to be exactly zero, the strike K must be less than or equal to this minimum possible price. Thus:

$$V_0^e = 0 \iff K \leq s_0(1+d)^T$$

Taking the negation of both sides yields the desired result: $V_0^e > 0$ if and only if $s_0 < \frac{K}{(1+d)^T}$.

Question 4: Positivity of the American Put

Problem Statement: Assume further that $-1 < d < 0 < r < u$. Show that $V_0 > 0$ if and only if $s_0 < \frac{K}{(1+d)^T}$.

Step-by-Step Solution:

1. **Sufficient Condition ($\Rightarrow$):** An American option gives the holder all the rights of a European option, plus the right to exercise early. Consequently, its value must be at least as great as the European equivalent: $V_t \geq V_t^e$ for all t. From Question 3, if $s_0 < \frac{K}{(1+d)^T}$, then $V_0^e > 0$. It immediately follows that $V_0 \geq V_0^e > 0$.

2. **Necessary Condition ($\Leftarrow$):** We must prove that if $s_0 \geq \frac{K}{(1+d)^T}$, then $V_0 = 0$. Suppose $K \leq s_0(1+d)^T$. Because we are now given that $d < 0$, the factor $(1+d)$ is less than 1. This implies that the sequence $(1+d)^t$ is strictly decreasing over time.

3. **Path Evaluation:** Since $(1+d)^t \geq (1+d)^T$ for all $t \leq T$, we have:

$$K \leq s_0(1+d)^T \leq s_0(1+d)^t$$

This means the strike price K is less than or equal to the minimum possible asset price at *every* time step t, not just at maturity.

4. **Conclusion:** Therefore, the immediate payoff $(K - S_t)_+ = 0$ holds Q-a.s. for all $t \in \{0, \ldots, T\}$. Since the option is never "in the money" at any point, its value V_0 must be 0.

Question 5: Evaluating Boundary Conditions

Problem Statement: Show that $v(0,0) = h(0)$ and then $v(0,K) > 0$.

Step-by-Step Solution:

1. **Evaluating at $S_0 = 0$:** We want to find the value of the American option if the underlying asset price drops to zero. If $S_t = 0$, the price remains stuck at zero because $0 \times (1+u) = 0$ and $0 \times (1+d) = 0$.

2. We calculate this using the recursive formulation from Question 1, starting at maturity T:

$$v(T,0) = h(0) = (K-0)_+ = K$$

Moving one step backward to $T-1$:

$$v(T-1,0) = \max\{h(0), \tilde{p}v(T,0) + \tilde{q}v(T,0)\}$$

Since $\tilde{p} + \tilde{q} = \frac{1-q}{1+r} + \frac{q}{1+r} = \frac{1}{1+r}$, we substitute to get:

$$v(T-1, 0) = \max\left\{h(0), \frac{h(0)}{1+r}\right\}$$

Because the interest rate $r > 0$, the discounted continuation value $\frac{h(0)}{1+r}$ is strictly less than the immediate exercise value $h(0)$. Therefore, the max is $h(0)$. Repeating this logic backward to $t = 0$ yields $v(0, 0) = h(0)$.

3. **Evaluating at $S_0 = K$:** We apply the condition derived in Question 4. We want to check if $V_0 > 0$ when $s_0 = K$. According to Question 4, $V_0 > 0$ if and only if $K < \frac{K}{(1+d)^T}$.

4. **Checking the Inequality:** Since $K > 0$ and $d < 0$, the term $(1+d)^T$ is strictly less than 1. Dividing K by a positive number less than 1 yields a result strictly greater than K. Therefore, the inequality $K < \frac{K}{(1+d)^T}$ holds true, which confirms that $v(0, K) > 0$.

3 The Early Exercise Boundary

Question 6: Existence of the Early Exercise Threshold

Problem Statement: Show that there exists $s_0^* \in [0, K)$ such that $v(0, x) = h(x)$ if $x \in [0, s_0^*]$ and $v(0, x) > h(x)$ if $x \in (s_0^*, \frac{K}{(1+d)^T})$.

Step-by-Step Solution:

1. **Defining the Threshold:** Let us define a candidate for our boundary point, s_0^*, as the supremum of all prices where the option value equals the immediate payoff for all values below it:

$$s_0^* := \sup\{x \in \mathbb{R}_+ : \forall y \in [0, x], v(0, y) = h(y)\}$$

2. **Existence and Bounds:** From Question 5, we know $v(0, 0) = h(0)$, meaning the set is non-empty and s_0^* exists. Also from Question 5, we established $v(0, K) > 0$. Since the payoff at strike is $h(K) = (K - K)_+ = 0$, we have $v(0, K) > h(K)$. Therefore, the threshold s_0^* must be strictly less than K, meaning $s_0^* \in [0, K)$.

3. **Equality at the Boundary:** Because h is a continuous function and the maximum operator preserves continuity (proved in Question 2), the value function $v(0, \cdot)$ is also continuous. Taking the limit as we approach the supremum implies $v(0, s_0^*) = h(s_0^*)$.

4. **Proving Equality on the Interval $[0, s_0^*]$:** We need to show the condition holds for any x in this interval. For any $x \in [0, s_0^*]$, we can express x as a convex combination of 0 and s_0^*. There exists an interpolation factor $l \in [0, 1]$ such that:

$$x = ls_0^* + (1 - l)0$$

Since we proved in Question 2 that $v(0, \cdot)$ is convex, we can apply the definition of convexity:

$$v(0, x) \leq lv(0, s_0^*) + (1 - l)v(0, 0)$$

Substituting the known equalities $v(0, s_0^*) = h(s_0^*)$ and $v(0, 0) = h(0)$:

$$v(0, x) \leq lh(s_0^*) + (1 - l)h(0)$$

The put payoff h is a straight line (affine) on the interval $[0, s_0^*]$ since $s_0^* < K$. Thus, the linear combination is exactly $h(x)$, giving us $v(0, x) \leq h(x)$. However, by the recursive formulation of the American option, $v(0, x) \geq h(x)$ is always true. Thus, $v(0, x) = h(x)$ for all $x \in [0, s_0^*]$.

5. **Proving Strict Inequality on $(s_0^*, \frac{K}{(1+d)^T})$:** We split this into two regions and evaluate them:

- *For $x \in (s_0^*, K)$:* By the very definition of the supremum s_0^*, any point slightly larger than s_0^* must break the equality condition. Thus, $v(0,x) > h(x)$.
- *For $x \in [K, \frac{K}{(1+d)^T})$:* From Question 4, we know $v(0,x) > 0$. Since $x \geq K$, the immediate payoff is $h(x) = (K-x)_+ = 0$. Hence, $v(0,x) > h(x) = 0$.

Combining these two facts, we see that $v(0,x) > h(x)$ holds strictly for the entire interval $(s_0^*, \frac{K}{(1+d)^T})$.

Question 7: Conclusion on the Infimum

Problem Statement: Conclude $s_0^* = \inf\{x \in \mathbb{R}_+ : v(0,x) > h(x)\}$.
Step-by-Step Solution:

1. **Connecting the Regions:** In Question 6, we perfectly partitioned the pricing domain below $\frac{K}{(1+d)^T}$ into two adjacent regions without any gaps:
   - The *immediate exercise region* $[0, s_0^*]$ where $v(0,x) = h(x)$.
   - The *continuation region* $(s_0^*, \frac{K}{(1+d)^T})$ where $v(0,x) > h(x)$.
2. **Defining the Infimum:** The mathematical set $\{x \in \mathbb{R}_+ : v(0,x) > h(x)\}$ represents the region where it is strictly optimal to hold the option rather than exercise it.
3. **Identifying the Boundary:** The lowest possible value (the infimum) of this strict inequality continuation region is exactly the upper bound (the supremum) of the equality exercise region. Therefore, it immediately follows that the threshold s_0^* is the infimum of the set of points where the option value is strictly greater than its intrinsic value.

4 Comparison with European Options

Question 8: Bounding the European Put

Problem Statement: For every $x \in [0, s_0^*]$, show $v^e(0,x) \leq h(x)$ and further $v^e(0,0) < h(0)$.
Step-by-Step Solution:

1. **General Inequality:** An American option provides the same rights as a European option, plus the additional right to exercise early. Because of this added flexibility, the value of the European option can never exceed the value of the American option. Therefore, for any x, we have $v^e(0,x) \leq v(0,x)$.
2. **Applying the Early Exercise Boundary:** In Question 6, we proved that for any $x \in [0, s_0^*]$, the American option value is exactly equal to its immediate exercise value: $v(0,x) = h(x)$. Substituting this into our general inequality gives $v^e(0,x) \leq h(x)$ for all $x \in [0, s_0^*]$.
3. **Evaluating at Zero:** We must show a strict inequality at $x = 0$. The initial value of the European put option is the discounted expected payoff at maturity:

$$v^e(0, s_0) = \frac{1}{(1+r)^T} \mathbb{E}_Q[h(S_T)]$$

If the initial price is $s_0 = 0$, the asset price remains at 0 until maturity, meaning $S_T = 0$ almost surely. The payoff at maturity is therefore exactly $h(0)$.

$$v^e(0,0) = \frac{h(0)}{(1+r)^T}$$

Because the interest rate $r > 0$, the discount factor $(1+r)^T$ is strictly greater than 1. Dividing $h(0)$ by a number strictly greater than 1 yields a result strictly less than $h(0)$. Thus, $v^e(0,0) < h(0)$.

Question 9: Ordering the Exercise Boundaries

Problem Statement: Let $s_0^e := \inf\{x \in \mathbb{R}_+ : v^e(0,x) > h(x)\}$. Show $s_0^* \leq s_0^e$.

Step-by-Step Solution:

1. **Understanding the Sets:** The point s_0^* is the threshold where the American option value strictly exceeds the payoff: $s_0^* = \inf\{x : v(0,x) > h(x)\}$. The point s_0^e is the analogous threshold for the European option: $s_0^e = \inf\{x : v^e(0,x) > h(x)\}$.

2. **Comparing the Values:** As established in Question 8, $v^e(0,x) \leq v(0,x)$ for all x. Because the European option value curve is always below or equal to the American option value curve, it takes a higher asset price for the European option to "break away" and become strictly greater than the intrinsic payoff line $h(x)$.

3. **Conclusion:** Since $v(0,x)$ reaches the strict inequality condition $v(0,x) > h(x)$ before or at the same time as $v^e(0,x)$ as x increases, the infimum of the American set must be less than or equal to the infimum of the European set. Hence, $s_0^* \leq s_0^e$.

Question 10: Option Price Graphs

Problem Statement: Draw the prices of American and European put options with respect to s_0.

Visual Representation: Below is a graph drawn using TikZ to illustrate the relationships proven in this TD.

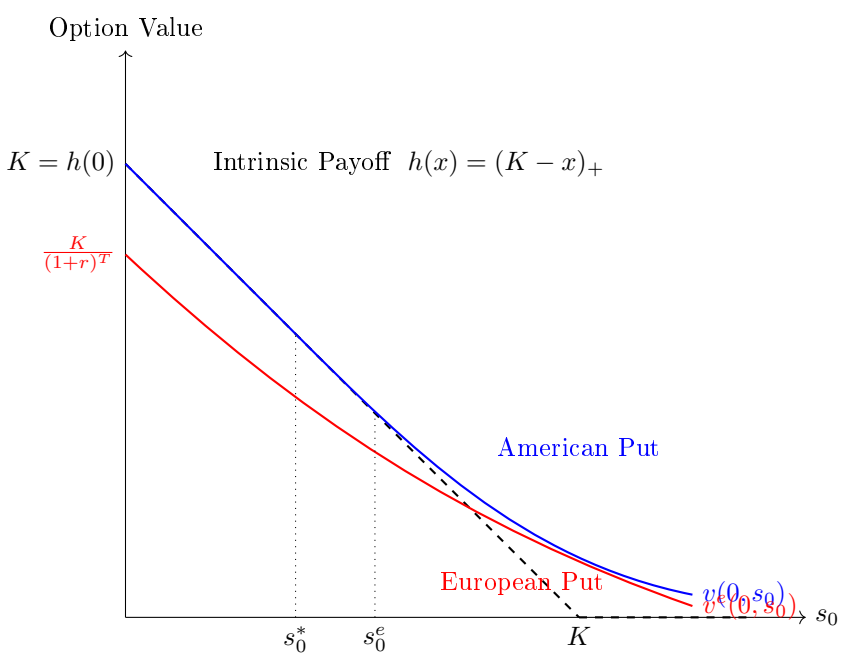

Figure 1: Prices of American and European put options with respect to the initial asset price s_0.